# Copy-Move Forgery Detection

**SYNOPSIS (Mini Project)**

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**COMPUTER SCIENCE AND ENGINEERING (AI & ML)**

***by***


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**UNDER SUPERVISION**

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# SYNOPSIS

**Team Details:**

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**Title of the Project**

Copy-Move Forgery Detection

**Brief Description of the Project**

In the digital era, image manipulation has become increasingly easy due to the availability of advanced editing tools. One of the most common forms of image tampering is Copy-Move Forgery (CMF), where a part of the image is copied and pasted within the same image to hide or duplicate content. Unlike other forgeries, CMF is difficult to detect because the copied region originates from the same image, thus sharing similar noise patterns, color palettes, and textures.

This project aims to develop a robust CMF detection pipeline that leverages feature extraction techniques (such as SIFT, ORB, or SURF) and traditional machine learning-based matching strategies (like k-d tree search, RANSAC) to identify duplicated regions. The pipeline will include:

1. Preprocessing for image normalization.
2. Feature extraction from keypoints.
3. Feature matching and geometric verification.
4. Forgery localization through post-processing (morphological operations, connected component analysis).

The system will be tested on benchmark datasets like CoMoFoD and COVERAGE, which provide forged and authentic images with ground truth masks. Quantitative metrics such as precision, recall, and Intersection over Union (IoU) will be used to evaluate the system's performance. The proposed solution will work without deep learning, making it transparent and interpretable while ensuring computational efficiency.

**Objectives**

1. To detect copy-move forgery in images using feature-based methods combined with geometric verification.
2. To accurately localize forged regions and evaluate detection performance using benchmark datasets.

**Existing state-of-art method/s in the proposed project**

Current CMF detection approaches fall into block-based and keypoint-based methods. Block-based methods, such as DCT or PCA on image blocks, are computationally expensive for high-resolution images and are sensitive to transformations. Keypoint-based methods (e.g., SIFT, SURF, ORB) extract distinct feature points and match them to detect duplication, offering robustness to transformations like scaling and rotation. Post-processing using algorithms like RANSAC helps eliminate false matches caused by repetitive patterns. While deep learning methods exist, they act as black boxes and are not allowed for this project; hence, the focus will be on transparent, traditional CV methods.

**Importance of the proposed project**

Copy-Move Forgery Detection plays a critical role in digital forensics, journalism, law enforcement, and intellectual property protection. By developing a solution based purely on interpretable CV and ML methods, this project provides a lightweight yet reliable tool for detecting tampered images, which can be integrated into forensic workflows or used in resource-limited environments where deep learning may be infeasible